6. Write short notes on the following
   i. Hofmann's bromide reaction
   ii. Ammonolysis
   iii. Gabriel phthalimide synthesis
   iv. Schotten – Baumann reaction
   v. Carbylamine reaction
   vi. Mustard oil reaction
   vii. Coupling reaction
   viii. Diazotisation
   ix. Gomberg reaction
7. How will you distinguish between primary secondary and tertiary alphatic amines.
8. Account for the following
   i. Aniline does not undergo Friedel – Crafts reaction
   ii. Diazonium salts of aromatic amines are more stable than those of aliphatic amines
   iii. pK_b of aniline is more than that of methylamine
   iv. Gabriel phthalimide synthesis is preferred for synthesising primary amines.
   v. Ethylamine is soluble in water whereas aniline is not
   vi. Amines are more basic than amides
   vii. Although amino group is o – and p – directing in aromatic electrophilic substitution reactions, aniline on nitration gives a substantial amount of m – nitroaniline.
9. Arrange the following
   i. In increasing order of solubility in water, $C_6H_5NH_2$,$(C_2H_5)_2NH$,$C_2H_5NH_2$
   ii. In increasing order of basic strength
   a) aniline, p- toludine and p – nitroaniline
   b) $C_6H_5NH_2$,$C_6H_5NHCH_3$,p-Cl-C_6H_4-NH_2
   iii. In decreasing order of basic strength in gas phase
   $(C_2H_5)NH_2$,$(C_2H_5)_2NH$,$\left(C_2H_5\right)_3$ N and NH_3
   iv. In increasing order of boiling point
   C_2H_5OH, $(CH_3)_2NH$, $C_2H_5NH_2$
   v. In decreasing order of the pK_b values
   $C_2H_5NH_2$, $C_6H_5NHCH_3$,$(C_2H_5)_2NH$ and CH_3NH_2
   vi. Increasing order of basic strength
   $C_6H_5NH_2$,$C_6H_5N(CH_3)_2$,$(C_2H_5)_2$ NH and CH_3 NH_2

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